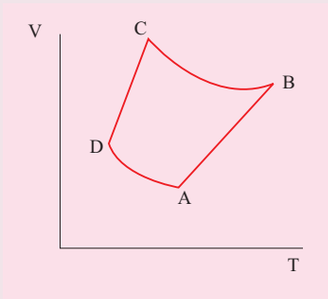
a) Process A

b) Process B

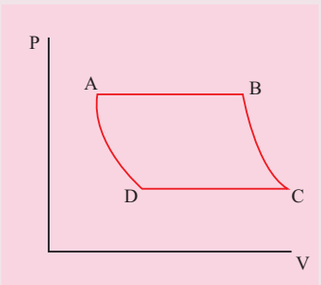
c) Process C

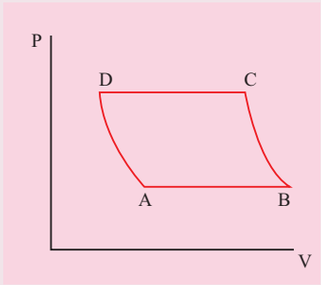
d) Equal work is done in Process A,B &C

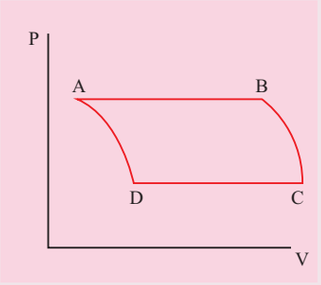
10. The V-T diagram of an ideal gas which goes through a reversible cycle $A \rightarrow B \rightarrow C \rightarrow D$ is shown below. (Processes $D \rightarrow A$ and $B \rightarrow C$ are adiabatic)

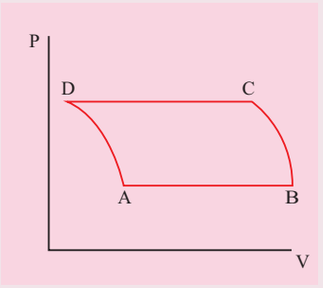


The corresponding PV diagram for the process is (all figures are schematic)

a)


b)


c)


d)


11. A distant star emits radiation with maximum intensity at 350 nm. The temperature of the star is

a) 8280 K b) 5000K

c) 7260 K d) 9044 K

12. Identify the state variables given here?

a) Q, T, W b) P, T, U

c) Q, W d) P, T, Q

13. In an isochoric process, we have

a) $W = 0$ b) $Q = 0$

c) $\Delta U = 0$ d) $\Delta T = 0$

14. The efficiency of a heat engine working between the freezing point and boiling point of water is

(NEET 2018)

a) 6.25% b) 20%

c) 26.8% d) 12.5%

15. An ideal refrigerator has a freezer at temperature −12°C. The coefficient of performance of the engine is 5. The temperature of the air (to which the heat ejected) is

a) 50°C b) 45.2°C

c) 40.2°C d) 37.5°C

Answers:

1) a	2) c	3) b	4) b
5) d	6) a	7) b	8) c
9) b	10) a	11) a	12) b
13) a	14) c	15) c	

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